Supplementary Material

Gain dependence in sample entropy profiling of electrohysterography and its effect on uterine-state associations

This supplement gives the feature definitions, the numerical controls, the complete gain-invariant (GI) associations and the robustness, amplitude and sensitivity analyses that support the Results. Exhaustive listings behind these tables are in the code repository (see Data and code availability in the main text).

Abbreviations. BH, Benjamini–Hochberg; CI, confidence interval; CR1, cluster-robust covariance with small-sample scaling; EHG, electrohysterography; FD, fractal dimension; FDR, false discovery rate; FM, fetal movement; FWH, fast-wave high-frequency; GI, gain-invariant; IN, guard-cleared inactivity; OLS, ordinary least squares; PSD, power spectral density; REML, restricted maximum likelihood; RMS, root mean square; SampEn, sample entropy; SD, standard deviation; SVD, singular-value decomposition; τ, embedding delay; UC, uterine contraction.

Table S1. Feature definitions and implementation parameters

The 34 original features of the fast-wave high-frequency (FWH) band are grouped by family below. The primary band was 0.3–1.0 Hz; signals were sampled at 200 Hz, filtered and decimated to 20 Hz, with 60-s windows. Welch settings were 10-s Hann segments, 5-s overlap, constant detrending and density scaling. Parameters not listed are fixed by the feature definition. Features were computed with the feature-engine package vendored in the repository (version 0.2.0.dev0; NumPy 2.3.5, SciPy 1.16.3). τ denotes the embedding delay.

Feature	Concise definition / formula	Units	Essential parameters	Citation
Time: median	sample median of x	signal units	none	
Time: kurtosis	Fisher excess kurtosis of x	dimensionless	bias=True	
Time: skewness	standardised third central moment	dimensionless	bias=True	
Time: mean absolute value	mean absolute value	signal units	none	
Time: RMS	square root of mean squared signal	signal units	none	(Hudgins et al., 1993)
Time: peak-to-peak	maximum minus minimum	signal units	none	
Time: waveform length	sum of absolute successive differences	signal units	sample differences	(Hudgins et al., 1993)
Time: zero-crossing count	number of sign changes between consecutive non-zero samples	count	exactly-zero samples removed first	
Time: slope sign-change count	number of interior points where adjacent slopes have opposite signs (product < 0)	count	slope threshold 0	
Frequency: peak frequency	frequency at maximum Welch PSD	Hz	Welch PSD	(Welch, 1967)
Frequency: mean frequency	power-weighted mean frequency	Hz	Welch PSD	(Welch, 1967)
Frequency: median frequency	frequency at cumulative half power	Hz	Welch PSD	(Welch, 1967)
Frequency: spectral edge (95%)	frequency at cumulative 95% power	Hz	Welch PSD	(Welch, 1967)
Frequency: spectral entropy	normalised Shannon entropy of PSD probabilities	dimensionless	Welch PSD	(Inouye et al., 1991)
Frequency: spectral spread	power-weighted frequency spread	Hz	Welch PSD	(Welch, 1967)
Entropy: approximate entropy	ApEn: phi(m)-phi(m+1)	dimensionless	m=2; r=0.2 population SD; Chebyshev; self matches included	(Pincus, 1991)
Entropy: permutation entropy	ordinal-pattern entropy	dimensionless	$m = 3$, $\tau = 1$	(Bandt and Pompe, 2002)

Feature	Concise definition / formula	Units	Essential parameters	Citation
Entropy: fuzzy entropy	$\ln \phi(m) - \ln \phi(m+1)$, where ϕ is the mean over distinct template pairs of $\exp[-\ln 2\,(d/r)^2]$ and d is the Chebyshev distance between mean-centred templates	dimensionless	m=2; delay=1; r=0.2 population SD; exponent=2	(Chen et al., 2007)
Entropy: distribution entropy	normalised Shannon entropy of the histogram of all pairwise Chebyshev distances between templates	dimensionless	m=2; delay=1; 500 equal-width bins from the smallest nearest-neighbour distance to the largest template range; divided by $\ln 500$	(Li et al., 2015)
Entropy: SVD entropy	entropy of singular-value distribution	dimensionless	$m = 3$, $\tau = 1$	(Roberts et al., 1999)
Entropy: fixed-r SampEn	negative log ratio of m+1 to m matches	dimensionless	m=2; r=0.2 population SD; Chebyshev; self matches excluded	(Richman and Moorman, 2000)
Entropy: Total SampEn	sum of retained SampEn-profile values	dimensionless	profile: m=2; aligned m/m+1 templates; distance rounded to 3 decimals; epsilon=1×10^{-12}	(Udhayakumar et al., 2018),(Udhayakumar, 2020)
Entropy: average SampEn	mean of retained SampEn-profile values	dimensionless	same profile settings as Total SampEn	(Udhayakumar et al., 2018),(Udhayakumar, 2020)
Entropy: median SampEn	median of retained SampEn-profile values	dimensionless	same profile settings as Total SampEn	(Udhayakumar et al., 2018),(Udhayakumar, 2020)
Entropy: profile SD	population SD of retained SampEn-profile values	dimensionless	ddof=0; same profile settings as Total SampEn	(Udhayakumar et al., 2018),(Udhayakumar, 2020)
Entropy: kurtosis SampEn	Fisher excess kurtosis of retained profile values	dimensionless	bias=True; same profile settings as Total SampEn	(Udhayakumar et al., 2018),(Udhayakumar, 2020)
Entropy: skewness SampEn	skewness of retained profile values	dimensionless	bias=True; same profile settings as Total SampEn	(Udhayakumar et al., 2018),(Udhayakumar, 2020)
Complexity: Lempel–Ziv	Lempel–Ziv sequence complexity	dimensionless	median binarisation; LZ76 phrase count; normalised by $\log_2(n)/n$	(Lempel and Ziv, 1976)
Complexity: Hjorth mobility	sqrt(variance of first derivative / variance of x)	dimensionless	none (closed form)	(Hjorth, 1970)
Complexity: Hjorth complexity	Hjorth complexity parameter	dimensionless	none (closed form)	(Hjorth, 1970)
Complexity: Fisher information	$\sum_i (p_{i+1} - p_i)^2 / p_i$ over the descending normalised singular values p of the mean-centred delay embedding	dimensionless	order 2, delay 1 (feature-engine defaults)	
Complexity: Katz fractal dimension	Katz fractal dimension	dimensionless	none (closed form)	(Katz, 1988)
Complexity: Higuchi fractal dimension	Higuchi fractal dimension	dimensionless	k_max=10	(Higuchi, 1988)
Complexity: detrended fluctuation analysis	DFA scaling exponent	dimensionless	first-order detrending; scales from 4 samples to one tenth of the window length, ratio 1.2	(Peng et al., 1994)

SampEn fixed-r uses m=2, Chebyshev distance, tolerance 0.2 times the population SD (ddof=0), aligned m/m+1 templates and self-match exclusion. The SampEn profile uses $m = 2$ and aligned templates but does not use a single proportional tolerance: its tolerance support is the set of occupied values of the pairwise Chebyshev distances after fixed three-decimal rounding; values are retained only when the shell is observed and both cumulative probabilities exceed 10^{-12}. Total and average are the sum and mean of retained values; median, SD, kurtosis and skewness use the implementation's population-SD / SciPy bias-default summaries. The rounded GI representation applies the same profile after division of the uncentred input by its population SD. The fixed-grid representation instead evaluates the dimensionless GI profile at a common grid without distance rounding.

Table S2. Original EHG9 UC–IN associations and estimator dependence

The mixed models use 42 women, 112 contributing recordings and 4,083 analysed windows (UC, IN and FM). Pooled OLS and woman-cluster CR1 use the 3,691 UC and IN windows (333 UC, 3,358 IN), standardised with the three-state mean and SD of the mixed model. The paired analysis uses the 34 women contributing both states. UC–IN denotes contraction minus guard-cleared inactivity (IN). Panel A gives the primary mixed-model coefficient (standardised; model-based Wald 95% CI) and the Benjamini–Hochberg (BH) q-value of each estimator within its own 34-feature family. Pooled OLS and woman-cluster CR1 share one coefficient and differ only in covariance. The paired woman-summary q-value belongs to a separate family. Panel B gives the intervals behind the main-text comparisons for the 11 features supported by the primary model; paired values are raw-unit median differences with percentile intervals from 5,000 resamples of women (not a whole-woman refit of the window-level model).

A. All 34 features

Feature	Mixed β	95% CI	q (mixed)	q (pooled OLS)	q (CR1)	q (paired)
Time: median	−0.2193	[−0.3289, −0.1097]	0.0005	0.0201	0.3193	0.1058
Time: kurtosis	0.0076	[−0.1066, 0.1218]	0.8959	0.1334	0.5533	0.1512
Time: skewness	0.1881	[0.0749, 0.3014]	0.0055	0.0058	0.0402	0.1988
Time: mean absolute value	0.3099	[0.1975, 0.4224]	1.12×10^{-6}	1.02×10^{-5}	0.4882	0.000176
Time: RMS	0.2691	[0.1566, 0.3817]	2.35×10^{-5}	0.000189	0.5147	0.000154
Time: peak-to-peak	0.2478	[0.1352, 0.3603]	0.000109	0.000657	0.5147	0.000154
Time: waveform length	0.3019	[0.1896, 0.4141]	1.55×10^{-6}	1.02×10^{-5}	0.4882	0.000176
Time: zero-crossing count	−0.0514	[−0.1293, 0.0264]	0.2806	0.0122	0.5533	0.6793
Time: slope sign-change count	−0.0236	[−0.099, 0.0518]	0.6552	0.0131	0.563	0.7148
Frequency: peak frequency	−0.0218	[−0.1124, 0.0688]	0.732	0.0034	0.3947	0.3307
Frequency: mean frequency	−0.0846	[−0.1662, −0.0029]	0.0899	0.0201	0.563	0.5329
Frequency: median frequency	−0.0852	[−0.1694, −0.00098]	0.0948	0.0143	0.5147	0.6498
Frequency: spectral edge (95%)	−0.1166	[−0.2027, −0.0305]	0.0269	0.6745	0.8795	0.3307
Frequency: spectral entropy	−0.0901	[−0.1933, 0.0132]	0.1648	0.552	0.8709	0.2294
Frequency: spectral spread	−0.1263	[−0.2076, −0.045]	0.0099	0.8061	0.9343	0.2286
Entropy: approximate entropy	−0.0092	[−0.1162, 0.0978]	0.8926	0.1033	0.563	0.6979
Entropy: permutation entropy	−0.0153	[−0.0929, 0.0622]	0.7457	0.0131	0.563	0.7976
Entropy: fuzzy entropy	−0.0572	[−0.1444, 0.0299]	0.2806	0.691	0.8795	0.5535
Entropy: distribution entropy	−0.0217	[−0.1328, 0.0894]	0.7457	0.0993	0.5147	0.5577
Entropy: SVD entropy	−0.0964	[−0.1766, −0.0161]	0.0528	0.0751	0.6553	0.5034
Entropy: fixed-r SampEn	−0.1154	[−0.2148, −0.016]	0.0558	0.0053	0.5147	0.3307
Entropy: Total SampEn	0.4092	[0.2959, 0.5225]	5.02×10^{-11}	2.21×10^{-10}	0.2745	0.000176
Entropy: average SampEn	−0.0602	[−0.1611, 0.0408]	0.3302	0.4173	0.7494	0.1414
Entropy: median SampEn	−0.0735	[−0.1832, 0.0362]	0.2806	0.5156	0.6736	0.9462
Entropy: profile SD	−0.0412	[−0.1375, 0.0551]	0.5061	0.4713	0.8235	0.2286
Entropy: kurtosis SampEn	0.0831	[−0.0303, 0.1965]	0.2649	0.0131	0.2166	0.1332
Entropy: skewness SampEn	0.0772	[−0.0327, 0.1871]	0.2725	0.0131	0.3193	0.1332
Complexity: Lempel–Ziv	0.0215	[−0.0702, 0.1132]	0.732	0.0122	0.5147	0.6457
Complexity: Hjorth mobility	−0.0915	[−0.1703, −0.0126]	0.0558	0.0306	0.5633	0.5034
Complexity: Hjorth complexity	−0.1435	[−0.2413, −0.0458]	0.0151	0.0026	0.2166	0.2286
Complexity: Fisher information	0.0896	[0.0099, 0.1693]	0.0624	0.032	0.5633	0.5034
Complexity: Katz fractal dimension	−0.055	[−0.1603, 0.0502]	0.3993	0.7299	0.8795	0.6457
Complexity: Higuchi fractal dimension	−0.09	[−0.1585, −0.0215]	0.0308	0.0765	0.6736	0.3307
Complexity: detrended fluctuation analysis	0.0618	[−0.0236, 0.1472]	0.2649	0.0201	0.563	0.5496

B. The 11 primary-supported features: intervals and woman-level agreement

Feature	Pooled/CR1 β	OLS 95% CI	CR1 95% CI	Paired median (raw)	Paired 95% CI	Women, same sign (%)
Time: median	−0.1456	[−0.2578, −0.0334]	[−0.2953, 0.0041]	-7.6×10^{-5}	$[-0.000219, 2.2 \times 10^{-5}]$	65
Time: skewness	0.1811	[0.069, 0.2931]	[0.0762, 0.2859]	0.032	[−0.0015, 0.058]	65
Time: mean absolute value	0.2962	[0.1782, 0.4142]	[−0.0901, 0.6825]	0.0027	[0.0019, 0.0035]	85
Time: RMS	0.2559	[0.1378, 0.3741]	[−0.1318, 0.6436]	0.0039	[0.0021, 0.0049]	85
Time: peak-to-peak	0.2353	[0.1171, 0.3535]	[−0.1522, 0.6229]	0.0221	[0.0107, 0.0297]	91
Time: waveform length	0.2977	[0.1797, 0.4157]	[−0.0876, 0.683]	0.5114	[0.3152, 0.6865]	82
Frequency: spectral edge (95%)	0.0288	[−0.0835, 0.1411]	[−0.2749, 0.3325]	−0.0123	[−0.0265, −0.0015]	68
Frequency: spectral spread	0.014	[−0.0976, 0.1255]	[−0.3261, 0.354]	−0.0038	[−0.0103, 0.0014]	62
Entropy: Total SampEn	0.4122	[0.2949, 0.5295]	[0.0366, 0.7879]	0.9821	[0.7865, 1.5475]	82
Complexity: Hjorth complexity	−0.2031	[−0.3167, −0.0895]	[−0.3693, −0.0368]	−0.0077	[−0.0132, −0.00066]	68
Complexity: Higuchi fractal dimension	0.1101	[−0.000844, 0.2211]	[−0.2335, 0.4537]	−0.0012	[−0.0044, 0.000332]	62

Table S3. Numerical validation controls

Gains were 0.25, 0.5, 1, 1.3, 2, 4 and 7.1 on 60 fixed windows (420 comparisons per feature). The time-domain features listed as degree 1 scale linearly with gain; every other listed feature has expected degree 0. A comparison passed when $|\text{computed} - \text{expected}| \leq 10^{-10} + 10^{-8}|\text{expected}|$; the same rule defined gain-1 reproduction and the terms "invariant" and "non-decreasing" in Table S7A.

A. Gain-1 reproduction and across-gain responses against expected laws

Feature group	Expected law	Comparisons per feature	Pass	Max abs. error
Gain-1 reproduction of stored reference values (34 features, 60 windows, 2,040 comparisons)	–	–	no failures	1.78×10^{-15}
Amplitude features (median, MAV, RMS, peak-to-peak, waveform length)	degree 1	420	all	7.11×10^{-15}
Other original features (23: remaining time, spectral, entropy and complexity features)	degree 0	420	all	1.07×10^{-14}
Entropy: Total SampEn	degree 0	420	60 of 420	72.7249
Entropy: average SampEn	degree 0	420	60 of 420	0.01
Entropy: median SampEn	degree 0	420	60 of 420	0.0033
Entropy: profile SD	degree 0	420	60 of 420	0.0686
Entropy: kurtosis SampEn	degree 0	420	60 of 420	132.4608
Entropy: skewness SampEn	degree 0	420	60 of 420	6.5729
GI Total and GI profile SD	degree 0	420	all	0

B. Full-precision control (12 windows) Total, average, profile SD and retained-support count were each unchanged across all gains (maximum difference 0; 12 of 12 windows with zero difference) when fixed-decimal distance rounding was removed without SD normalisation.

C. Decimal-resolution ablation Same 60 windows and six non-unit gains (0.25, 0.5, 1.3, 2, 4, 7.1); Total, average and SD SampEn. Differences are relative to the unit-gain value of the same summary. "Worst median difference" is the largest absolute median relative difference over windows, and "worst maximum difference" the largest absolute maximum, each taken over gains and summaries. "Pass proportion" is the fraction of window–gain–summary comparisons meeting the tolerance above. "Approx. expected bins" is the approximate number of occupied rounding bins expected for that resolution.

Resolution	Mode	Approx. expected bins	Measured median bins	Worst median difference (%)	Worst maximum difference (%)	Pass proportion
Original, 3 decimals	Original	80	84.0	630.6386	673.1571	0.0
Original, 4 decimals	Original	800	793.0	604.1487	629.0289	0.0
Original, 5 decimals	Original	7,000	6902.5	604.2731	610.6597	0.0
GI, 1 decimal	GI	60	60.5	0	0	1.0
GI, 2 decimals	GI	600	588.0	0	0	1.0
GI, 3 decimals	GI	5,000	5258.0	0	0	1.0

D. Independent implementation of the fixed-grid profile (12 windows) An explicit pairwise Chebyshev implementation matched the reference implementation after matching template starting indices, for normalised area under the profile, slope, centroid, profile SD, kurtosis, skewness and the valid fraction: maximum absolute difference 0 for every summary, with validity masks in agreement. This is an internal implementation check, not a replication, and it does not test agreement of state associations between representations. The fixed-grid check used 200 dimensionless tolerance points spanning 0.01–4.00 The required valid-point fraction was 0.90; the observed minimum across 300 IN windows was 1.0.

Table S4. Complete gain-invariant state associations

The common population is 42 women, 112 contributing recordings and 4,083 analysed windows. Coefficients are model-standardised window-level woman/recording mixed-model estimates with model-based Wald 95% CIs. The final declared 16-test Benjamini–Hochberg family spans four channels, two GI summaries and two contrasts and was selected after development. The 13-feature development family is identified in Table S5B.

Channel	Summary	Contrast	Estimate	95% CI	Raw p	Final q (16-test)
EHG9	GI Total	FM–IN	−0.2003	[−0.2878, −0.1129]	7.16×10^{-6}	1.43×10^{-5}
EHG9	GI Total	UC–IN	−0.1131	[−0.2034, −0.0227]	0.0142	0.0206
EHG9	GI profile SD	FM–IN	−0.246	[−0.3381, −0.1539]	1.65×10^{-7}	3.77×10^{-7}
EHG9	GI profile SD	UC–IN	−0.1398	[−0.235, −0.0447]	0.004	0.0071
EHG10	GI Total	FM–IN	−0.2736	[−0.3601, −0.187]	5.85×10^{-10}	2.34×10^{-9}
EHG10	GI Total	UC–IN	−0.0896	[−0.179, −0.000266]	0.0493	0.0658
EHG10	GI profile SD	FM–IN	−0.3452	[−0.4373, −0.2531]	2.04×10^{-13}	3.26×10^{-12}
EHG10	GI profile SD	UC–IN	−0.1297	[−0.2248, −0.0345]	0.0076	0.0121
EHG11	GI Total	FM–IN	−0.2937	[−0.3836, −0.2038]	1.51×10^{-10}	8.06×10^{-10}
EHG11	GI Total	UC–IN	−0.0466	[−0.1395, 0.0462]	0.3252	0.3469
EHG11	GI profile SD	FM–IN	−0.3323	[−0.4278, −0.2368]	8.99×10^{-12}	7.19×10^{-11}
EHG11	GI profile SD	UC–IN	−0.0621	[−0.1608, 0.0366]	0.2172	0.2673
EHG12	GI Total	FM–IN	−0.2545	[−0.345, −0.164]	3.57×10^{-8}	9.52×10^{-8}
EHG12	GI Total	UC–IN	−0.0439	[−0.1374, 0.0497]	0.3579	0.3579
EHG12	GI profile SD	FM–IN	−0.2787	[−0.3753, −0.1821]	1.57×10^{-8}	5.04×10^{-8}
EHG12	GI profile SD	UC–IN	−0.0538	[−0.1538, 0.0461]	0.2913	0.3329

Table S5. Exclusion scenarios and analysis-family map

S5A. Annotation and whole-woman influence sensitivity

E0 is the full analysed dataset. E1 removes the five retained windows carrying movement, equipment-manipulation or position-change flags. E2 removes only the flagged ice037 window. E3 removes all ice037 windows. Each scenario reports the model structure/status and the scenario-specific feature SD used for standardisation. Coefficients are not comparable as raw physiological magnitudes across scenarios because the denominators change; interval exclusion is not treated as FDR-supported robustness.

Feature	Scenario	Women	Windows	Estimate	95% CI	Feature SD	Model / status
Time: RMS	E0	42	4,083	0.2691	[0.1566, 0.3817]	0.086	woman-only; woman-only boundary
GI Total	E0	42	4,083	−0.1131	[−0.2034, −0.0227]	51.1195	woman/recording nested; nested clean
GI profile SD	E0	42	4,083	−0.1398	[−0.235, −0.0447]	0.0218	woman/recording nested; nested clean
Time: RMS	E1	42	4,078	0.6734	[0.5813, 0.7656]	0.0075	woman/recording nested; nested clean
GI Total	E1	42	4,078	−0.1039	[−0.1944, −0.0134]	50.9838	woman/recording nested; nested clean
GI profile SD	E1	42	4,078	−0.1289	[−0.2242, −0.0335]	0.0218	woman/recording nested; nested clean
Time: RMS	E2	42	4,082	0.6742	[0.5823, 0.766]	0.0076	woman/recording nested; nested clean
GI Total	E2	42	4,082	−0.0993	[−0.1897, −0.009]	50.97	woman/recording nested; nested clean
GI profile SD	E2	42	4,082	−0.1236	[−0.2188, −0.0284]	0.0218	woman/recording nested; nested clean
Time: RMS	E3	41	3,992	0.7015	[0.6096, 0.7934]	0.0075	woman/recording nested; nested clean
GI Total	E3	41	3,992	−0.1069	[−0.1978, −0.016]	50.9594	woman/recording nested; nested clean
GI profile SD	E3	41	3,992	−0.1301	[−0.226, −0.0342]	0.0218	woman/recording nested; nested clean

S5B. Analysis and multiplicity map

This map records the reporting role and selection timing for analyses discussed in the manuscript. Post-development choices are not described as prespecified.

Analysis	Population / channel	Contrast	Estimator	Feature family	Adj. size	Role	Timing
Original EHG9 features	EHG9	UC–IN	woman/recording mixed model	34-feature	34	primary original-feature analysis	defined before final modelling; not preregistered
Model dependence	EHG9	UC–IN	pooled OLS; woman-cluster CR1; paired woman summary	34-feature / paired summary	34 per family	estimand comparison	secondary; not one covariance correction
Gain-law audit	60 windows	positive gains 0.25-−7.1	descriptive per-window trajectories	none	none	numerical mechanism	after development
Two-summary GI	EHG9	UC–IN and FM–IN	woman/recording mixed model	secondary two-summary	2 per contrast	secondary GI family	after development

Analysis	Population / channel	Contrast	Estimator	Feature family	Adj. size	Role	Timing
Cross-channel GI	EHG9–EHG12	UC–IN and FM–IN	woman/recording mixed model	final cross-channel BH	16	exploratory GI inference	selected after development
GI development family	EHG9	UC–IN and FM–IN	woman/recording mixed model	13-feature GI development family	13 per contrast	screening; GI Total and GI profile SD carried forward	development
Conditional analyses	EHG9	UC–IN and FM–IN	mixed models with log RMS or shape covariates; leave-one-woman-out refits	none	none	exploratory conditioning, outside the declared family	after the primary analysis
Annotation/influence sensitivity	EHG9	UC–IN	scenario-specific mixed-model fits	descriptive	none	interpretation boundary	named scenarios

Table S6. Robustness and residual diagnostics for the 11 primary EHG9 UC–IN features

Panels A–C are descriptive and do not form a second inferential family. Coefficients are standardised UC-minus-IN estimates; the 11 features are those supported by the primary EHG9 mixed model in the 34-feature family.

S6A. Lag-1 autocorrelation of model residuals in the five most-windowed EHG9 recordings

Residuals were ordered by window start time within each recording (ice002_p_1of3, ice044_p_2of3, ice004_p_1of1 with 58 windows each, ice003_p_1of2 with 56 and ice036_p_1of6 with 55). Across the 11 features and five recordings, 25 of 55 combinations reached the informal $|r_1| \geq 0.30$ flag, with a maximum $|r_1|$ of 0.560. For 20 of the 55 combinations (the four amplitude features whose primary fit was a woman-only boundary fit) the random-effect variance was estimated at zero, so conditional and marginal residuals coincide. This is a diagnostic, not a model of serial correlation.

Feature	Recordings flagged (of 5)	Max $\lvert r_1 \rvert$
Median	0	0.221
Skewness	1	0.307
MAV	4	0.528
RMS	4	0.524
Peak-to-peak	2	0.433
Waveform length	3	0.560
Spectral edge 95	2	0.445
Spectral spread	1	0.420
Total SampEn	4	0.532
Hjorth complexity	0	0.191
Higuchi FD	4	0.553

S6B. Balanced IN subsampling, window duration and clean-UC sensitivity

Balanced IN: 1,000 within-woman resamples of IN windows (same-direction and BH-supported frequencies, and the median ratio of the resampled to the primary-fit standard error). Duration: window length 40, 60 and 80 s with the same event centres. Clean UC: 21 retained UC windows overlapping an FM-centred window removed (312 UC windows).

Feature	Same dir.	q<0.05 freq.	SE ratio	40 s β	40 s q	80 s β	80 s q	Clean β	Clean q
Median	1.000	0.245	1.41	−0.288	6.7e-05	−0.203	2.5e-04	−0.219	8.3e-04
Skewness	1.000	0.488	1.36	0.188	0.003	0.182	0.010	0.180	0.012
MAV	1.000	0.000	3.35	0.392	2.0e-08	0.268	1.6e-05	0.323	8.2e-07
RMS	1.000	0.000	3.36	0.308	8.3e-07	0.245	2.1e-04	0.283	1.5e-05
Peak-to-peak	1.000	0.000	3.36	0.231	3.8e-06	0.258	6.1e-04	0.262	6.7e-05
Waveform length	1.000	0.000	3.35	0.253	8.4e-08	0.346	1.9e-05	0.316	1.0e-06
Spectral edge 95	1.000	0.282	1.26	−0.111	0.047	−0.120	0.013	−0.103	0.067
Spectral spread	1.000	0.283	1.30	−0.106	0.047	−0.128	0.005	−0.117	0.026
Total SampEn	1.000	0.002	3.21	0.523	3.8e-13	0.352	8.9e-09	0.422	2.7e-11
Hjorth complexity	1.000	0.075	1.35	−0.083	0.206	−0.145	0.007	−0.137	0.029
Higuchi FD	1.000	0.251	1.26	−0.114	0.013	−0.093	0.014	−0.087	0.054

S6C. Conditional-residual distribution and spread

Descriptive only (no formal test). Max/min SD ratio is the largest ratio of residual SD between states.

Feature	Skewness	Excess kurtosis	Max/min SD ratio
Median	0.57	6.85	1.24
Skewness	1.78	24.46	1.11
MAV	61.87	3910.84	27.01
RMS	62.90	3997.65	42.35
Peak-to-peak	63.04	4010.00	47.38
Waveform length	61.97	3919.03	28.18
Spectral edge 95	−0.04	1.05	1.02
Spectral spread	0.21	0.91	1.04
Total SampEn	58.79	3653.19	16.49
Hjorth complexity	0.21	1.04	1.13
Higuchi FD	0.56	2.02	1.04

Table S7. Amplitude dependence of the entropy associations

S7A. Direct positive scaling of 60 windows (20 per state)

Median relative change from unit scale (%). “Non-decr.” is the number of the 60 trajectories that were non-decreasing across scales but not invariant (the two categories are exclusive); “Invariant” is the number numerically unchanged at every scale.

Summary	0.25×	0.5×	2×	4×	Non-decr.	Invariant
RMS	-75.0	-50.0	+100.0	+300.0	60	0
Total SampEn	-75.4	-50.4	+102.3	+309.2	60	0
Average SampEn	-5.6	-1.7	+2.1	+4.6	55	0

Summary	0.25×	0.5×	2×	4×	Non-decr.	Invariant
Median SampEn	-12.1	-4.2	+4.2	+8.8	32	0
Fixed-r SampEn	+0.0	+0.0	+0.0	+0.0	0	60

S7B. Four-channel UC/IN models with and without standardised log-RMS

Pooled UC/IN windows (3,691 physical windows, 14,764 channel rows; 42 women; EHG9 reference channel). Mixed: woman random intercept plus physical-window variance component, maximum likelihood. CR1: fixed-effects OLS with woman-cluster CR1 covariance. Outcomes and log-RMS standardised over the pooled UC/IN population.

Outcome	Estimator	Model	UC β	95% CI	p	log-RMS β
Total SampEn	Mixed (ML)	unadjusted	0.299	[0.216, 0.383]	2.3e-12	–
Total SampEn	Mixed (ML)	adjusted	−0.086	[−0.166, −0.005]	0.036	0.543
Total SampEn	OLS, woman-cluster CR1	unadjusted	0.297	[0.023, 0.571]	0.034	–
Total SampEn	OLS, woman-cluster CR1	adjusted	0.030	[−0.075, 0.134]	0.570	0.374
fixed-r SampEn	Mixed (ML)	unadjusted	−0.213	[−0.307, −0.119]	8.4e-06	–
fixed-r SampEn	Mixed (ML)	adjusted	0.235	[0.150, 0.321]	7.4e-08	−0.646
fixed-r SampEn	OLS, woman-cluster CR1	unadjusted	−0.149	[−0.402, 0.104]	0.241	–
fixed-r SampEn	OLS, woman-cluster CR1	adjusted	0.083	[−0.129, 0.295]	0.435	−0.325
SVD entropy	Mixed (ML)	unadjusted	−0.053	[−0.135, 0.030]	0.209	–
SVD entropy	Mixed (ML)	adjusted	0.020	[−0.061, 0.101]	0.630	−0.105
SVD entropy	OLS, woman-cluster CR1	unadjusted	0.153	[−0.167, 0.473]	0.339	–
SVD entropy	OLS, woman-cluster CR1	adjusted	0.256	[0.027, 0.485]	0.029	−0.144

Residual diagnostics for the adjusted Total SampEn mixed model: the Spearman correlation between the conditional residual and log RMS was −0.70 (pooled; −0.69 to −0.73 across the four channels), against −0.98 for marginal residuals. Across channels, the Spearman correlation between each woman's UC-minus-IN change in Total SampEn and her change in RMS was 0.92–0.96 (34 paired women; 0.79–0.98 for the other amplitude summaries).

S7C. EHG9 UC-minus-IN coefficients for original and GI Total SampEn under log-RMS conditioning, with and without the extreme window

Three-state random-intercept model (woman and recording-within-woman), REML, 4,083 windows; custom block-diagonal REML implementation, checked against the reference unadjusted coefficients (agreement to four decimals). Log-RMS is standardised. Cross-checked against statsmodels 0.14.5 MixedLM (REML, woman intercept plus recording variance component): maximum absolute coefficient difference 3×10^{-6}, maximum standard-error difference 0.003. The second window set removes the one UC window from woman ice037 (samples 6,094–18,094; RMS 5.49 against 0.17 for the next largest window); outcomes are re-standardised within each window set.

Outcome	Adjustment	Window set	UC β (SE)	95% CI	FM β	log-RMS β
Original Total	none	all 4,083	0.409 (0.058)	[0.296, 0.522]	0.078	–
Original Total	none	4,082	0.692 (0.038)	[0.618, 0.765]	0.203	–
Original Total	standardised log-RMS	all 4,083	0.020 (0.054)	[−0.086, 0.126]	−0.122	0.529
Original Total	standardised log-RMS	4,082	0.203 (0.021)	[0.161, 0.245]	−0.062	0.749
GI Total	none	all 4,083	−0.113 (0.046)	[−0.203, −0.023]	−0.200	–
GI Total	none	4,082	−0.099 (0.046)	[−0.190, −0.009]	−0.199	–
GI Total	standardised log-RMS	all 4,083	0.275 (0.040)	[0.196, 0.354]	0.002	−0.567
GI Total	standardised log-RMS	4,082	0.280 (0.040)	[0.201, 0.359]	0.009	−0.579

S7D. Leave-one-woman-out refits (42 refits, each dropping one woman)

UC-minus-IN coefficient	Minimum	Median	Maximum
GI Total, unadjusted	−0.134	−0.113	−0.088
GI Total, log-RMS adjusted	0.244	0.275	0.305
Original Total, unadjusted	0.213	0.411	0.442
Original Total, log-RMS adjusted	0.005	0.019	0.059

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